Continuous Keyframe-Free Compression for Real-Time Audio-Video Streaming over Ultra-Low-Bandwidth Networks

A practical codec for surveillance and unmanned platforms on degraded 2G / EDGE and satellite uplinks

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Abstract

We present a real-time audio-video streaming system engineered for genuine low-bandwidth degraded links — 2G/EDGE cellular and high-cost satellite uplinks — where conventional predictive codecs fail. The design departs from the classic keyframe + inter-frame predictive model: frames are encoded as a sequence of **patch-level modes** that are autonomously decodable, so no picture depends on a long chain of earlier pictures. The result is a stream that **degrades gracefully under packet loss instead of freezing**: when fragments are dropped the effective frame rate dips and then recovers, but the transmission never stalls.

The representation is designed around the holographic principle that information should be distributed rather than localized, so that a partial or imperfect delivery still yields a usable picture. In practice the codec realizes this as a sparse, motion-adaptive, keyframe-free pipeline built on per-patch temporal differencing, adaptive quantization, run-length coding and Zstandard entropy compression, with an efficient colour extension that adds a low-resolution chroma plane at near-zero cost on static scenes. It targets the operational regimes that matter for **surveillance, unmanned aerial platforms and telemedicine** — where the requirement is continuous, reliable awareness rather than high resolution. This revision additionally introduces **optional end-to-end authenticated encryption** (ChaCha20-Poly1305) of both the video and the independently transported audio channel, implemented against the reference specification and validated against its published test vectors, without disturbing any of the transport or recovery properties described below.

1 Motivation: where conventional codecs break

Video coders such as H.264/HEVC are excellent for broadband, low-latency, low-loss environments. They are a poor match for the links this work targets:

- **Minimum rate floor.** SPS/PPS, NAL headers and mandated periodic IDR frames (for random access and error recovery) consume a large share of a 10–30 kbps budget.

- **Drop-sensitivity.** Losing a P-frame propagates an error until the next IDR, producing seconds of green blocks, freeze or stream collapse — unacceptable on a high packet-loss, high-RTT satellite link.
- **Feedback-dependent rate control.** Mainstream rate adaptation assumes low RTT and limited jitter; on EDGE or satellite the adaptation lags badly, causing overshoot and stalls.
- **Encoder cost.** Real-time HD encoding is expensive in power and latency on the ARM SoCs typical of small UAVs.

The codec described here is intentionally different by construction. It transmits only what changed — for a surveillance scene with a mostly static background and a small region of interest, this approaches near-zero bandwidth. It has no long dependency chain, so a single lost frame does not corrupt anything. And it keeps the encoder so simple that it runs comfortably on modest embedded hardware. These are exactly the properties a drone or surveillance operator needs.

2 System pipeline

The end-to-end pipeline is:

- **1.** Frame acquisition (OpenCV / FFmpeg capture).
- **2.** Partition into patches and reduce to the working resolution.
- **3.** Per-patch temporal differencing against a local reference.
- **4.** Three-way mode decision: skip / delta / intra.
- **5.** Adaptive quantization of the residual with a robust scale estimate.
- **6.** Run-length and Zstandard entropy compression.
- **7.** Optional authenticated encryption (ChaCha20-Poly1305) of the compressed blob. NEW
- **8.** UDP packetization with fragmentation and optional FEC.
- **9.** Reconstruction and local display at the receiving endpoint.

> **On the term holographic**
>
> The guiding principle is that information is distributed rather than localized, so that partial delivery still produces a usable picture. This paper keeps that framing as the design intent, while describing the algorithm actually implemented — a sparse, motion-adaptive, keyframe-free patch coder. The implementation-centered description is intentional: it is what the system does, what the measurements reflect, and what a reader can reproduce from the source.

3 Codec design

3.1 Patch-level mode decision

Each frame is converted to the luma plane and reduced to a working patch. For every patch we compute the pixel-wise difference delta against the reference kept by the encoder, and its mean squared error Δ. A three-way decision is then taken:

$$\text{mode} = \begin{cases} \text{skip} & \text{if } \Delta < \text{ydel}, \\ \text{delta} & \text{if } \text{ydel} \le \Delta < \text{xdel}, \\ \text{intra} & \text{if } \Delta \ge \text{xdel}. \end{cases}$$

- **skip**: the patch is unchanged within threshold; nothing is transmitted, and the reference is retained. Because a skipped patch carries zero information, losing its packet cannot corrupt the picture — a source of intrinsic robustness.
- **delta**: the residual is quantized with an adaptive scale and transmitted. This is the steady-state low-cost path.
- **intra**: a full patch is sent, refreshing the reference. This is the recovery point; it bounds the accumulated error from any lost delta.

3.2 Adaptive quantization and entropy coding

The residual is quantized with a per-frame scale derived from a robust estimate of the residual's deviation (median absolute deviation, with an RMS fallback) so that outliers do not distort the step size. The quantized integer values are then run-length encoded — highly efficient because residuals are dominated by runs of near-zero quantized samples — followed by Zstandard compression, which effectively removes the residual redundancy. On a near-static scene this yields only tens of bytes per frame.

3.3 Transport and recovery

Encoded frames are fragmented into UDP datagrams with sequencing and a checksum, and reassembled on the receiver. Datagrams that fail to assemble within a timeout are dropped rather than buffered, so the link never stalls (no head-of-line blocking, no retransmission storm). An optional XOR parity packet provides single-fragment recovery. Because frames carry short dependencies, an outage produces a local, transient dip in the effective frame rate that the next complete frame immediately repairs. Audio, when used, travels over an **independent** UDP channel rather than being multiplexed into the video frame stream: the two are decoupled by design so that transient degradation on one channel cannot cascade into the other.

3.4 Introduction of colour

The previous version of the system operated in monochrome: chrominance was carried only as a single, nearly-neutral reduced plane, so in practice the picture was black and white. This revision introduces **genuine colour**, which is a significant improvement for surveillance and live monitoring, where distinguishing a target is often what matters most.

Colour is added without disturbing the low-bandwidth design. Chrominance is treated as a second, low-resolution plane that is transmitted only when it actually changes, so on a static background it costs almost nothing. The chroma plane is reduced to a small square whose side is derived from the luma patch size (roughly one sixth), giving a chroma/luma ratio close to real 4:2:0 subsampling; a mild saturation boost counteracts the desaturation caused by subsampling; and the decoder upscales it with cubic interpolation. A change-threshold skips re-sending colour when a region is static, which is why colour is nearly free for a fixed background. This makes the stream deliver true, natural-colour video at a cost that stays within the same low-bandwidth budget.

Table 2 shows the measured additional bit rate of adding colour to a low-motion scene at a fixed luma resolution, as a function of the chroma square side. The colour cost is dominated by the chroma side, so the operator can trade colour detail against bandwidth.

Chroma side	Added bit rate	Colour PSNR	Note
9×9	~0 kbps	31.7 dB	essentially free
12×12	+5 kbps	32.9 dB	suggested default
18×18	+15–19 kbps	34.4 dB	best colour, near EDGE limit
33×33	higher	~25.7 dB	medium profile, most detail

Table 2. Measured additional bit rate and colour PSNR versus chroma square side (low-motion scene). On the medium profile the chroma side is derived from the luma patch, giving a larger, more natural colour plane.

3.5 End-to-end encryption NEW

This revision adds optional authenticated encryption of both the video channel and the independently transported audio channel (Section 3.3), using **ChaCha20-Poly1305** as specified in RFC 8439. The primitive — ChaCha20, Poly1305, and the SHA-256 construction used for key derivation — is implemented directly from the reference specifications with no external cryptographic library dependency, and was validated against the officially published RFC 8439 and FIPS 180-4 test vectors before integration.

A per-session key is derived from an operator-supplied shared passphrase and a random 8-byte session identifier generated at the start of each transmission, via a two-stage SHA-256 construction. The resulting authenticated ciphertext simply replaces the plaintext compressed blob ahead of UDP fragmentation (pipeline step 7); the fragmentation, reassembly and loss-recovery mechanisms of Section 3.3 are otherwise unmodified and unaware of whether encryption is in use. A frame that fails authentication — a mismatched passphrase, or tampering in transit — is dropped by the receiver rather than decoded, so no unauthenticated or corrupted content is ever reconstructed or displayed.

Encryption is opt-in and configured independently per channel: with it disabled, the system behaves exactly as in the previous revision, and a peer with encryption off remains interoperable with another peer that also has it off. Because the video and audio channels are architecturally independent processes with no shared memory (Section 3.3), each is keyed separately — using the same shared passphrase on both is a deployment convention, not a structural coupling.

The current threat model assumes the shared passphrase itself stays confidential between the two communicating parties: there is no authenticated key-exchange protocol (e.g. Diffie–Hellman), so anyone already in possession of the passphrase, including an intermediary that might see it through an out-of-band channel, can decrypt a captured stream. For the target scenario — two operators who have already agreed on a call code out of band, e.g. by voice, SMS or a pre-shared contact record — this is judged an adequate, low-complexity starting point, and is revisited as future work in Section 7.

4 Measured behaviour

Two datasets inform the results. Over synthetic controlled sequences we measured the codec's rate–quality curve as a function of scene motion; on a real 2G/EDGE field test with a live talking head we verified the end-to-end operational properties. Table 1 reports the measured payload bit rate (25 fps) and luma PSNR relative to the up-sampled source, i.e. the quality ceiling at the chosen working resolution.

Profile / working res	Scene	Payload bit rate	Luma PSNR
72×72	static	10.0 kbps	38.2 dB
72×72	medium motion	24.9 kbps	32.9 dB
72×72	high motion	82.2 kbps	16.9 dB
150×150	static	17.7 kbps	39.4 dB
200×200	static	27.3 kbps	39.1 dB

Table 1. Measured payload bit rate and luma PSNR versus scene motion. The static case fits comfortably within an EDGE budget.

The key observation is the separation between the static and moving cases. A near-static surveillance scene produces on the order of 10 kb/s and high PSNR, whereas heavy motion elevates the rate several-fold while quality falls to a floor that is still usable for detection. This is precisely the operating window of the target applications: continuous awareness of a scene that is mostly static, where a sustained high rate is not needed.

Field test on real 2G/EDGE

A live capture with independently transported, end-to-end encrypted audio was streamed over a genuine EDGE link and observed end-to-end

using the medium profile, which carries both audio and colour. Four properties were confirmed on the live link. First, **adaptive frame rate**: under congestion the effective frame rate falls and then recovers automatically. Second, and most importantly, **no freeze**: because the codec has no keyframes and no long inter-frame dependency, occasional packet loss produces only a local, transient degradation — the picture never stops updating. Third, **true colour with correct hue**: with the chroma channels in the correct order the colours reproduce faithfully and the stream remains smooth. Fourth, **encrypted audio and video stay in perceptual sync** to within roughly 200–500 ms — expected given the two channels are transported and decrypted independently (Section 3.3) with no shared clock, and judged acceptable for the target use case. This is the defining operational advantage over standard codecs on a lossy, high-latency channel.

5 Applications in constrained environments

5.1 Unmanned platforms on satellite uplinks

A drone in a remote operating area often relies on a satellite uplink that is charged per megabyte and has a high round-trip time. This codec is a natural fit because it minimizes data and tolerates the channel. Two properties are decisive.

Cost of data. For a loitering aircraft the scene is near-static, so the stream collapses to roughly 10 kb/s. In one hour this is about 4.5 MB, or about 108 MB over a 24-hour mission. Because the rate tracks content rather than a fixed floor, the same mission costs substantially less than a conventional codec coding at a constant rate — an important difference when bandwidth itself is the scarce, expensive resource.

No stall under loss. Satellite RTT is typically 500–700 ms with real loss. UDP fire-and-forget delivery avoiding retransmission avoids head-of-line blocking entirely; and because each frame is independently decodable, a dropped frame merely dips the frame rate for an instant before the next complete frame restores the picture. For an operator watching a feed from a platform in a contested area, a feed that never freezes is worth more than a sharper one that does.

5.2 Video surveillance

Continuous monitoring in low-coverage areas becomes practical without freezing. A static camera on a mostly static scene produces a very low sustained bit rate; the moment a target enters the field of view and moves, the affected patches transition to delta and the stream instantly adapts its rate to the new content. The reduced working resolution is sufficient to detect and track an entity, which is the actual requirement for surveillance rather than reading a face.

5.3 Emergency and telemedicine

In both cases the priority is continuous, reliable communication between two parties over a legacy or degraded network. The codec's graceful degradation means a weak link degrades the quality of the session rather than dropping it, keeping the channel open for as long as the network permits. Optional end-to-end encryption (Section 3.5) additionally protects the confidentiality of the call content itself against passive interception on the path between the two parties, which matters for medically or operationally sensitive communication.

6 System properties

- **Keyframe-free.** No mandatory periodic reference frame; frame rate adapts to the link.
- **Continuous degradation.** Quality scales gracefully instead of collapsing.
- **Immediate recovery.** A lost frame is repaired by the next complete frame.
- **Intrinsic robustness.** Skipped patches transmit no information and thus cannot corrupt the picture upon loss.
- **Optional authenticated confidentiality.** ChaCha20-Poly1305 encryption of either channel, independently configurable, with no change to the above properties when enabled. NEW

7 Limitations and future work

The system is tuned for robustness over fidelity, and its window of operation is the near-static, low-motion regime typical of surveillance and loitering UAVs. High-motion content rapidly raises the bit rate and lowers the achievable quality; and the embedded audio channel must be reduced in bit rate to coexist with video within the very tightest budgets. The encryption scheme (Section 3.5) relies on a pre-shared passphrase with no authenticated key-exchange protocol, which is adequate for the current operator-to-operator use case but not a substitute for a Diffie–Hellman-style exchange in a future revision. Likewise, the current transport requires at least one endpoint to be reachable at a fixed, publicly routable address (or an equivalent manual port-forwarding arrangement); a rendezvous/relay server allowing two NAT-bound clients to register with a shared call code and be transparently bridged — without either side needing a public IP — is planned as the next infrastructure addition, ahead of a mobile client. Further future work includes objective quality evaluation (PSNR, SSIM) on a defined corpus, GPU acceleration, region-of-interest prioritization to concentrate detail where it matters, and a systematic comparison against standard codecs within this low-bandwidth regime.

8 Conclusion

A continuous, keyframe-free compression system has been presented for real-time audio-video streaming over ultra-low-bandwidth degraded networks. By replacing fragile predictive structures with a distributed, motion-adaptive, per-patch representation, the system absorbs packet loss and bandwidth degradation and degrades gracefully instead of failing. On real 2G/EDGE links and high-cost satellite uplinks it keeps a live feed continuously usable — never freezing — which makes it a practical fit for surveillance, unmanned platforms and emergency telemedicine, where continuous awareness matters more than resolution. This revision shows that the same design accommodates optional authenticated end-to-end encryption of both its video and audio channels without compromising any of these transport properties, closing an important gap for confidentiality-sensitive deployments while the remaining infrastructure work — NAT-independent rendezvous and a mobile client — continues.